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Archival Test Paper 1

INTRODUCTION

Institutions are built upon our taken-for-granted assumptions (Berger and Luckmann, 1967). These assumptions silently guide our everyday behavior (Zucker, 1977), making our social interactions more automatic and predictable (DiMaggio and Powell, 1991) and our prevailing institutional arrangements more self-activating and resilient (Jepperson, 1991). As a result, these taken-for-granted assumptions are said to be subtlest yet most powerful force underlying the reproduction of our institutions (Suchman, 1995). Yet while assumptions often go without saying, this does not mean they are never mentioned. This is especially true for leaders who are tasked with safeguarding the stability of our social institutions (Kraatz, 2009), and regularly reaffirm these assumptions to reinforce their influence. Indeed, corporate executives often elaborate upon the assumptions of their strategic business plans (Drucker, 1994), professionals reassert the assumptions that define jurisdictional boundaries (Suddaby and Greenwood, 2005), and top government officials continually discuss the assumptions that ground their policymaking (Plosser, 2012). What happens to the stability of an institution when a prominent leader explicitly reaffirms these taken-for-granted assumptions?

Research on the micro-foundations of institutions sheds some light on this question. Scholars in this space are interested in the micro-level processes that influence institutions (Powell and Rerup, 2016) and, in particular, how individual actors can play a role in this by tinkering with these underlying assumptions (Lawrence and Suddaby, 2006). Although much of this research has focused on how actors deliberately try to disrupt these assumptions to change institutions (Seo and Creed, 2002; Munir and Phillips, 2005), more recent work has shown that actors also attempt to reinforce these assumptions to maintain the existing social order (Heaphy, 2013). For example, scholars have demonstrated how leaders often engage in rituals (Dacin, Munir, and Tracey, 2010) or storytelling activities (Zilber, 2009) to help institutionalized meanings persist over time. However, while these

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forms of maintenance work bolster and reproduce the prevailing institution, they appear to do so without explicitly engaging its underlying assumptions. As a result, we still know little about the institutional consequences of leaders who try to reinforce the existing social order by simply reaffirming these assumptions directly.

This study proposes that a careful examination of these considerations reveals strong theoretical reasons to believe that explicitly reaffirming these assumptions will actually disrupt, not reproduce, the prevailing institution. I suggest that to understand why this is, we need to delve more deeply into the nature of taken-for-grantedness. When institutional assumptions achieve a taken-for-granted status, they take on an objective, natural, or fact-like quality (Berger and Luckmann, 1967; Garfinkel, 1967). As this happens, these assumptions get detached "from the presumed control of the very actors who initially created them" (Suchman, 1995: 583), obscuring their contingent and social origins (Schutz, 1967; Douglas, 1986) and making alternative ways of behaving literally unthinkable (Zucker, 1983). Yet when assumptions reach this point, Suchman warns, "any overt attention—including supportive attention—may have the detrimental side effect of disrupting taken-for-grantedness" (1995: 596). Indeed, talking openly about these assumptions, even if it is to reaffirm them, will bring them to mind. Just thinking about them reopens their obscured and contingent origins (Garfinkel, 1967), making them appear less objective or natural and revealing to people "that there are other possible, attractive alternatives" (Zucker, 1977: 728) to the current way of doing things. This line of reasoning suggests that explicitly reaffirming our taken-for-granted assumptions will destabilize the prevailing institution.

To develop this idea, I integrate our understanding of the taken-for-grantedness of institutions with research on linguistic arguments. In particular, I draw on Toulmin's (1958) model of argument—which includes the four components of *data*, *warrant*, *claim*, and *backing*—to propose that the way speakers structure their arguments maps onto the taken-for-granted structure of our social institutions (see Harmon, Green, and Goodnight, 2015). Specifically, speakers often make arguments using *data* and *warrants* to convince others of particular *claims*, all of which are grounded in collectively

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understood assumptions—or *backing*—that provide the consensual basis upon which people conduct their ongoing interactions. A key insight of this study is that Toulmin's linguistic concept of backing can be mapped directly onto institutional theorists' concept of taken-for-granted assumptions. This insight, as we shall soon see, is important because it enables me to theorize and empirically examine the possibility that explicitly reaffirming these taken-for-granted assumptions—or rather, the *backing* of an argument—will inadvertently destabilize the prevailing institution.

I test this idea in the context of the Federal Reserve, an institution sitting at the epicenter of the United States financial system that, if destabilized, creates uncertainty in the broader financial markets. This empirical setting is particularly noteworthy because the argument I am proposing runs counter to the prevailing wisdom of central bankers (Brandvold and Halvorsen, 2011). Indeed, it is well-documented that central bankers believe that reaffirming the assumptions underlying their monetary policy framework provides more transparency into the inner workings of their historically secretive institution (Yellen, 2013; Bernanke, 2015), and that this should reinforce the strength of their institution and reduce market uncertainty, not increase it (Plosser, 2012). Thus, given the practical importance of this predicted market outcome, I also explore several factors that might mitigate or amplify this presumably unintended effect. I do so by leveraging research that argues emotions shape how audiences make sense of important or uncertain events in financial markets (Abolafia and Kilduff, 1988; Pfarrer, Pollock, and Rindova, 2010). A synthesis between this work and the emotions the Fed has focused on historically led me to explore how two factors—the positive tone of Fed speeches and the level of fear in the business media—shape the way the market interprets the Fed Chair's discussion of these assumptions. Taken together, this paper integrates research on arguments and emotions with the micro-foundations of institutions in order to provide a novel perspective on the complex relationship between language and institutions.

ARGUMENTS, EMOTIONS, AND INSTITUTIONS

Language and Institutions

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Institutional and organization theorists have long recognized that there is a close connection between language and institutions. Berger and Luckmann (1967), for instance, famously discussed how institutions are built upon language. The idea is that institutions somehow get codified in the way people talk (Garfinkel, 1967; Schutz, 1967), with some meanings naturally becoming more legitimate than others. Since people tend to gravitate towards things that are more legitimate (DiMaggio and Powell, 1991; Suchman, 1995), they more readily use these legitimate meanings, thereby perpetuating the very institution in which they reside (Zucker, 1977). This story, however, resides at a fairly general level. As a result, theorists ever since have tried to find more concrete ways to understand how the key characteristics of our social institutions relate to the language we use in everyday life.

The most common approach has been to conceptualize institutions as systems of statements or words that cohere together—often referred to as vocabularies or discourses. Meyer and Rowan (1977), for instance, argued that vocabularies form rationalized accounts that generate isomorphic pressures to conform. Similarly, Kunda (1992) conceptualized a strong culture of normative control as being based partly on the continued use of certain words that reflected and reinforced the prevailing culture. Scholars have also demonstrated that people can use vocabularies and discourses to influence institutions (Suddaby and Greenwood, 2005; Vaara, 2014), and that changes in these systems of words are often associated with known changes in prevailing institutions (Fiss and Hirsch, 2005; Ocasio and Joseph, 2005; Colyvas and Powell, 2006). Research in this area is still flourishing too. Recent reviews have been published on the vocabularies (Loewenstein, Ocasio, and Jones, 2012) and discourse perspectives (Phillips and Oswick, 2012), and these ideas are featured prominently in the still-growing institutional logics perspective (Thornton, Ocasio, and Lounsbury, 2012) and a recent special issue on the communicative foundations of institutions (Cornelissen et al., 2015).

Yet one of the most important characteristics of an institution is that some meanings are more taken-for-granted than others (Zucker, 1977; Jepperson, 1991). Indeed, scholars define institutions as "taken-for-granted repetitive social behavior that is underpinned by normative systems and cognitive

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understandings that give meaning to social exchange and thus enable self-reproducing social order" (Greenwood et al., 2008: 4), and some have even highlighted this "taken-for-granted" element as the very essence of an institution (Phillips and Malhotra, 2017: 400). However, while the aforementioned perspectives on vocabularies and discourses excel at explaining how language hangs together in a coherent system, they are not as useful in identifying the extent to which these meanings are taken-for-granted. What we need, I propose, is a new theoretical perspective that enables us to connect the vocabularies or discourses we use in everyday life to this taken-for-granted structure of institutions. I suggest that one way to do this is to examine the underlying structural components of arguments.

Arguments and Institutions

Arguments are a way of reasoning with others. Stephen Toulmin, an esteemed British philosopher, established one of the most authoritative ways to analyze how people use arguments (Toulmin, 1958; Toulmin, Rieke, and Janik, 1984). This approach—known as the Toulmin Model (see Figure 1)—suggests that all arguments contain at least four components: *data*, *warrant*, *claim*, and *backing*. At the most basic level, people reason with others by asserting a *claim* (i.e., conclusion), and then justifying this claim with *data* (i.e., evidence) and *warrants* (i.e., explanations for why the data support the claim). However, Toulmin also recognized a critical fourth component. Specifically, he pointed out that this basic level of reasoning is always grounded upon collectively understood assumptions—or *backing*—that provide the "rules of the game" for how people interact. This observation leads to an important insight. Although people often leave the backing implicit and talk *within* the rules of the game, they can also make the backing explicit and talk *about* the rules of the game themselves (Goodnight, 1993; Harmon, Green, and Goodnight, 2015).

[Insert Figure 1 about here]

Consider the game of American football, specifically in the context of the National Football League (NFL). The NFL has many rules, such as when to use instant replay, which penalties can be called, and even the etiquette its players should observe. Now consider the public commentary of the

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league commissioner, the person tasked with safeguarding the integrity of the NFL. Often times, the commissioner will not discuss these rules directly and, instead, focus on discussing the activities occurring within the game itself. For instance, he may talk about how the penalty called last week was correct or how a player is being fined for flagrant behavior, often elaborating upon these claims with more data to explain what happened. However, sometimes the commissioner will step back and publicly discuss the rules themselves as a way to reassert their influence. He may, for instance, rearticulate the wording of the rule, clarify its intended purpose, discuss its origins, and so on. In fact, leaders often shift in their everyday talk between leaving the backing implicit (i.e., talking *within* the rules of the game) and discussing the backing explicitly (i.e., talking *about* the rules of the game).

Now consider the context of the Fed, where the game is not football but United States monetary policymaking, and the commentator is not the NLF commissioner but the Fed Chair. The rules or backing underlying this game are the assumptions related to the monetary policy framework of the Fed, which specifically concern the objectives and conventional tools used to conduct central banking activities in the United States. The Fed's objectives, which are also known as their dual Congressional mandate, are to maximize employment and maintain price stability. To achieve these objectives, the Fed has long used a variety of conventional tools, like engaging in open market operations, setting the discount rate, and changing member bank reserve requirements. This framework is largely taken as a given (Abolafia, 2010), having remained reasonably stable in its current form since the Federal Reserve Reform Act of 1977. These rules of the game thus make up the taken-for-granted assumptions underlying a bounded, specialized discourse on United States monetary policymaking, the express purpose of which is to stabilize the broader financial system ("The Fed - FAQs," 2017).

Just as in the NFL commissioner example, public speeches given by the Fed Chair can exhibit similar variation in how explicit they make these assumptions. For instance, some speeches reason more *within* the rules of the game, making claims about the state of the U.S. economy and providing economic data to support these claims. For example, in a speech to the Economic Club of Washington,

D.C., on December 7, 2009, Chair Bernanke asserts an initial claim about the recent recovery of the economy and then provides three pieces of data to justify this claim:

> A number of factors support the view that the recovery will continue next year (*claim*). Importantly, corporations are having relatively little difficulty raising funds in the bond and stock markets (*data*), stock prices and other asset values have recovered significantly from their lows (*data*), and a variety of indicators suggest that fears of systemic collapse have receded substantially (*data*) (Bernanke, 2009).

In contrast, in speeches where the Chair is talking more *about* the rules of the game, he or she explicitly reaffirms the backing, further clarifying the nature or boundaries of monetary policy. For example, in a speech given at the Federal Reserve Bank of St. Louis on October 11, 2001, Chair Greenspan makes explicit the assumptions underlying the Federal Reserve System:

> We at the Federal Reserve are given two mandates that are not often spelled out explicitly. First, to implement an effective monetary policy to meet our legislated objectives. Second, to do so in a most open and transparent manner in recognition that we, as unelected officials, are accountable both to the Congress from which we derive our monetary policy mission and to the American people (*backing*) (Greenspan, 2001).

What I am proposing is that Toulmin's linguistic concept of *backing* can be mapped directly onto institutional theorists' concept of taken-for-granted assumptions. Specifically, the backing is quite literally the linguistic articulation of an institution's taken-for-granted assumptions. While the backing may not be entirely taken-for-granted, the key is that the backing is at least more taken-for-granted than the other argument components and, because of this, may have important implications when made explicit. These implications may be particularly relevant for leaders who, as noted earlier, regularly reaffirm these assumptions as one way to reinforce the prevailing institution.

Arguments and Market Uncertainty

If the Fed Chair explicitly reaffirming these taken-for-granted assumptions does, in fact, reinforce the prevailing institution, then we would expect such efforts to reduce uncertainty in the broader financial markets. This expectation seems to be consistent with the belief of central bankers

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(Yellen, 2013; Bernanke, 2015; "The Fed - FAQs," 2017), who presume that such efforts will help bolster confidence in their institution and clarify the implications of their actions for the markets. However, I argue that opposite is true. In particular, I contend that explicitly reaffirming these taken-for-granted assumptions does not reassert their influence, but instead, disrupts the natural course of things, destabilizing the institution and increasing overall market uncertainty.

To understand why, first consider what happens when the Fed Chair does not discuss the assumptions or backing underlying their monetary policy framework. The more the Fed Chair leaves the backing implicit, the more it should reinforce the market's taken-for-grantedness of these assumptions. Since the backing is already collectively understood by the market, the Fed Chair not discussing it only strengthens the cognitive legitimacy of these ideas and the notion that they do indeed "go without saying." By reinforcing their objective and fact-like status, the Chair further obscures the contingent and social origins or these assumptions (Schutz, 1967; Douglas, 1986) and makes alternative ways of conducting monetary policy and influencing the market more unthinkable (Zucker, 1983). The result of these assumptions remaining intact as taken-for-granted facts is that it encourages individual market participants to continue deferring as they normally would to "the way one always does things," thereby leading to continued "conformity and isomorphism" in judgments and decisions made in the market (Bitektine and Haack, 2015: 53-54; Suchman, 1995; Harmon, Green, and Goodnight, 2015). As a result, speeches containing lower amounts of backing will continue to reinforce the prevailing monetary policy framework and constrict the overall range of expected directions the market could go in the future, thereby reducing uncertainty in the market.

However, the more the Fed Chair explicitly reaffirms the backing, the more this will diminish the market's taken-for-grantedness of these assumptions. Given that the backing typically goes without saying, the Chair bringing these assumptions to mind prompts market participants to think about or at least consider the foundations of this framework (Bitektine and Haack, 2015: 58; Harmon, Green, and Goodnight, 2015). The implication of the Chair bringing these assumptions to mind, even if done

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subtly, is that it removes some of the original status of these assumptions as taken-for-granted facts (Suchman, 1995), thereby revealing to people "that there are other possible, attractive alternatives" (Zucker, 1977: 728) to the current way of conducting monetary policy. As Garfinkel explains, once someone modifies "the objective structure of the familiar, known-in-common environment by rendering the background expectancies inoperative," this forces individuals to try and "manage the reconstruction of the natural facts by himself [or herself] and without consensual validation" (1967: 54). Thus, diminishing the market's taken-for-grantedness of these assumptions will reduce market participants' ability to rely on "the way one always does things" to make forecasts about the future, producing greater heterogeneity in overall judgments and decisions made in the market (Thessaly, 2008). As a result, the more a speech explicitly reaffirms the backing, the more this will destabilize the prevailing monetary policy framework and expand the overall range of expected directions the market could go in the future, thereby increasing market uncertainty.

> **Hypothesis 1 (H1):** The more a Fed Chair's speech explicitly reaffirms the backing underlying the monetary policy framework, the more market uncertainty will increase.

Emotions and Market Uncertainty

When important or uncertain events occur in financial markets (e.g., when the Fed Chair reaffirms these assumptions), emotions have been shown to influence how market participants interpret the meaning of such events (Abolafia and Kilduff, 1988). This growing body of work has identified two sources of emotion that meaningfully effect this process: the emotion used by the speaker (Green, 2004; Suddaby and Greenwood, 2005) and the emotion expressed in the broader media (Pollock and Rindova, 2003; Pfarrer, Pollock, and Rindova, 2010). I consider both of these sources in the context of Fed, paying careful attention to the specific emotions the Fed itself has focused on historically when communicating with the market. My argument is that emotions—specifically, the positive tone of the Fed Chair's speech and the fear expressed in the business media—create an interpretative context that

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shapes how market participants collectively make sense of the Fed Chair's discussion of these assumptions.

Speech Positive Tone. Across a variety of contexts, research has demonstrated the benefits of communicating in a positive tone. For instance, information conveyed in a positive light leads audiences to rate people's performances as better (Levin, 1987), perceive management control systems as stronger (Schneider, Holstrum, and Marden, 1993), and view organizational practices as more favorable (Davis and Bobko, 1986). These benefits also appear to hold when organizations share important information with the financial market. Davis, Piger, and Sedor (2012) show that using a positive tone in earnings press releases produces a better short-term stock market reaction, and that this result holds even after controlling for firm characteristics related to fundamentals (Huang, Teoh, and Zhang, 2013). These studies suggest that conveying important information in an overall positive tone focuses market participant's attention on more optimistic interpretations of this announcement, thereby creating greater confidence and certainty about the future.

In the context of the Fed, market participants similarly pay attention to the level of positivity in the Chair's communications (Cruikshank and Sicilia, 1999; Holmes, 2013). However, I suggest that while the overall positive tone of a speech may indeed influence market evaluations directly, more importantly, it may also function as an interpretive context that shapes how market participants collectively make sense of specific ideas that the Fed Chair is discussing (Pfarrer, Pollock, and Rindova, 2010). Recall that Hypothesis 1 contends that when the Fed Chair begins to reaffirm the backing explicitly, this will prompt market participants to think of alternatives to the current way of doing things and lead them to make decisions in increasingly diverging directions. When this is done within the context of a speech that has an overall more positive tone, I argue that this positivity will constrict market participant's attention to the more optimistic alternatives. What this means is that the range of expected directions the market could go in the future, which originally expanded as the Fed Chair talked more about the backing, will now be narrowed due to the higher levels of positive

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emotion. Thus, the more positive tone a Fed Chair speech contains, the more this will suppress the uncertainty created from discussing the backing.

Hypothesis 2 (H2): The positive tone of the speech will suppress the market uncertainty created from the Fed Chair explicitly reaffirming the backing underlying the monetary policy framework.

Business Media Fear. While the emotion used by a speaker can shape how market participants interpret important and uncertain events, so too can the emotion expressed in the broader business media. This ongoing media narrative contributes to what some emotions scholars have called an emotional climate (Menges and Kilduff, 2015), which can be understood as the prevailing zeitgeist of the market that influences the attention of market participants. Pfarrer, Pollock, and Rindova (2010), for instance, show how the emotion expressed about a firm in the media can shape how investors interpret and react to earnings announcements. In context of the Fed, the primary emotion they have historically been most concerned about is fear (Krugman, 2001; Holmes, 2013). Indeed, fear can create a more pessimistic emotional climate that can influence how market participants interpret events (Lerner and Keltner, 2001; Lerner, Small, and Loewenstein, 2004). For this reason, fear is widely acknowledged as the primary driver of financial panics (Shiller, 1988, 2000; Holmes, 2013; Bernanke, 2015), the very market phenomenon the Fed was established as an institution to avoid.

High levels of fear in the business media may create an especially problematic context within which to discuss these taken-for-granted assumptions. I propose that the mechanism at play here is identical to the mechanism for positive tone, except that it works in the opposite direction. That is, while high levels of positive tone create a more optimistic interpretative context, high levels of fear create a more pessimistic interpretive context within which market participants make sense of market events. As before, we begin with the baseline thesis that the Fed Chair explicitly reaffirming the backing will prompt market participants to think of alternatives to the current way of doing things, leading them to make decisions in increasingly diverging directions. If there are high levels of fear in

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the business media at the time of the speech, this particularly pessimistic emotional climate will exacerbate this divergence. What this means is that the range of expected directions the market could go in the future, which originally expanded as the Fed Chair talked more about the backing, will now be expanded even more because they were discussed in an emotional climate of fear. As a result, while discussing the backing within the context of an overall more positive speech will suppress uncertainty, discussing the backing within the context of a more fearful emotional climate will amplify uncertainty.

Hypothesis 3 (H3): The fear expressed in the business media at the time of the speech will amplify the market uncertainty created from the Fed Chair explicitly reaffirming the backing underlying the monetary policy framework.

METHODS

Sample

My initial sample consisted of all Fed Chair speeches given between January 1, 1998 and December 31, 2014, totaling 344. Seven speeches were removed because, based on an outlier analysis, their studentized residuals exceeded plus or minus three. The final sample thus consisted of 337 speeches (159 by Alan Greenspan, 166 by Ben Bernanke, and 12 by Janet Yellen) (see Figure 2).

[Insert Figure 2 about here]

Endogeneity

In the ideal experiment, I would be able to randomly assign the amount of backing made explicit in Fed speeches and observe the change in the level of market uncertainty on the day of the speech. Since I am unable to perform this ideal experiment, market factors that influence changes in market uncertainty may also influence how the Fed talks. To address this concern, I not only control for a number of market factors, but I also conducted interviews with former Fed Chair Ben Bernanke and former Fed Governors Donald Kohn and Mark Olson to gain additional comfort over the speechwriting process. Key insights from these interviews as well as several additional empirical tests

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are discussed at the end of the results section in order to provide further evidence of the robustness of my main findings.

Dependent Variable

I measure *market uncertainty* using the change in the VIX volatility index. The VIX is a daily index calculated by the Chicago Board Options Exchange "by averaging the weighted prices of S&P 500 puts and calls over a wide range of strike prices" (Chicago Board Options Exchange, 2003: 2). The VIX thus represents options traders' estimates of the direction of the S&P 500 over the next month by providing an aggregate measure of the variance of option prices on any given day. The higher the variance, the more uncertainty there is in the market. The VIX is the standard approach finance scholars use to measure market uncertainty (Connolly, Stivers, and Sun, 2005; Ang et al., 2006; Andersson, Krylova, and Vähämaa, 2008; Bia'lkowski, Gottschalk, and Wisniewski, 2008). Moreover, consistent with research that examines the effect of Fed communication on the VIX (Nikkinen and Sahlström, 2004; Chen and Clements, 2007; Nikkinen, Sahlström, and Äijö, 2007; Jamali, 2009; Vähämaa and Äijö, 2011), I use a two-day event window (t_{-1} to t_0) to measure the change in market uncertainty on the day of the speech. I identified the speech date from the Federal Reserve website, and validated this with data received from my Freedom of Information Act Request No. G-2015-00191.

Independent Variables

Backing Ratio (BR). The backing ratio measures the relative amount of backing made explicit by the Fed Chair in each speech. The author and three business school undergraduate students familiar with economics and monetary policy coded each paragraph of each speech as one that either discusses the backing or does not discuss the backing. I calculated interrater reliability at the paragraph level (Neuendorf, 2001; Krippendorff, 2003) for the first 60 speeches (Krippendorff's alpha = 0.88) as well as for the last 10 speeches (Krippendorff's alpha = .84) to demonstrate consistency. I then used this paragraph-level coding scheme to calculate the backing ratio for each speech:

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BR = (number of paragraphs that make the backing explicit / total number of paragraphs)

For a paragraph to be coded as not making the backing explicit, the Chair must engage *only* the structural components of data or warrants to make claims. However, they need not engage in all three components. Most frequently, the Chair provided some sort of evidence about actions they have taken or economic-related metrics they have collected in order to make a claim about the state of the economy. For instance, in a speech on September 26, 2005 to the American Bankers Association, Chair Greenspan makes an initial claim, supports this claim with data, and then reasserts the claim. Keep in mind that the argument components of data, warrants, and claims were not coded. I provide them here only to clarify what a typical "no backing" paragraph looks like in this context.

> This enormous increase in housing values and mortgage debt has been spurred by the decline in mortgage interest rates, which remain historically low (*claim*). Indeed, the thirty-year fixed-rate mortgage, currently around 5 3/4 percent, is about 1/2 percentage point below its level of late spring 2004, just before the Federal Open Market Committee (FOMC) embarked on the current cycle of policy tightening (*data*). This decline in mortgage rates and other long-term interest rates in the context of a concurrent rise in the federal funds rate is without precedent in recent U.S. experience (*claim*) (Greenspan, 2005).

Similarly, in a speech on January 3, 2014 to the American Economic Association, Chair Bernanke makes an initial claim and then supports this claim with a variety of data.

> The economy has made considerable progress since the recovery officially began some four and a half years ago (*claim*). Payroll employment has risen by 7-1/2 million jobs (*data*). Real GDP has grown in 16 of 17 quarters (*data*), and the level of real GDP in the third quarter of 2013 was 5-1/2 percent above its pre-recession peak (*data*). The unemployment rate has fallen from 10 percent in the fall of 2009 to 7 percent recently (*data*). Industrial production and equipment investment have matched or exceeded pre-recession peaks (*data*) (Bernanke, 2014).

For a paragraph to be coded as making the backing explicit, the Chair at some point must explicitly and directly talk about the backing. This generally occurs when the Chair reaffirms or reiterates in some way the nature and boundaries of monetary policy objectives and conventional tools. For instance, in a speech on October 18, 2011 at the Federal Reserve Bank of Boston's 56th Economic

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Conference, Chair Bernanke reaffirms the Fed's dual Congressional mandate, followed by a clarification of how inflation targeting fits into their monetary policy framework.

> The Federal Reserve is accountable to the Congress for two objectives—maximum employment and price stability (*backing*), on an equal footing—and it does not have a formal, numerical inflation target. But, as a practical matter, the Federal Reserve's policy framework has many of the elements of flexible inflation targeting. In particular, like flexible inflation targeters, the FOMC is committed to stabilizing inflation over the medium run while retaining the flexibility to help offset cyclical fluctuations in economic activity and employment (Bernanke, 2011).

Similarly, in her speech on March 5, 2014 after being sworn in as 15th Chair of the Federal Reserve, Chair Yellen reaffirms the core objectives of the Fed and then reiterates her commitment towards achieving them.

> The goals set by Congress for the Federal Reserve are clear: maximum employment and stable prices (*backing*). It is equally clear that the economy continues to operate considerably short of these objectives. I promise to do all that I can, working with my fellow policymakers, to achieve the very important goals Congress has assigned to the Federal Reserve (*backing*) (Yellen, 2014).

Speech positive tone. Following existing work (Pfarrer, Pollock, and Rindova, 2010; Rhee and Fiss, 2014), I used text analysis software Linguistic Inquiry and Word Count (LIWC) to create an index that captures the relative amount of positive emotional words in the speech. Positive emotional content is captured using a dictionary approach, whereby the percentage of words psychometrically related to positive emotion (e.g., happy, good, nice, positive, great, favorable, etc.) are reported in relation to all words in a speech. This positive emotion word dictionary was compiled and validated by Pennebaker and his colleagues (Pennebaker, Booth, and Francis, 2007; Pennebaker et al., 2007).

Business media fear. I was granted full access to the Thomson Reuters Market Psych Indices proprietary database, which calculates a daily measure of the relative level of fear expressed in the United States business media. Every five minutes, their algorithms scrape business news media sources (e.g., Reuters, Wall Street Journal, Financial Times) using a dictionary approach, reporting a daily measure of the amount of words appearing in these sources psychometrically related to fear (e.g.,

worrisome, concerning, anxious, fearful, panicky, etc.) in relation to all words in the business media discourse. In order to capture the context of fear within which market participants interpret the Fed Chair's speech, I constructed a business media fear variable by taking the average of this fear index over a three-day window (t_{-1} to t_{+1}) around the day of the speech.

Control Variables

Market-related factors. I controlled for market conditions that could simultaneously influence the BR of speeches as well as changes in market uncertainty. I controlled for *existing market uncertainty* prior to the speech by calculating the 30-day average VIX before the day of the speech. I controlled for the prior month's *unemployment rate* and *inflation rate* because they are market indicators of the Fed's operations and performance. I controlled for *expansionary* and *contractionary* monetary policy conditions by creating two dummy variables that were coded "1" if the most recent FOMC meeting resulted in lowering the federal funds rate (i.e., expansionary monetary policy) or raising the federal funds rate (i.e., contractionary monetary policy). Since the level of dissent in FOMC meetings might reflect underlying issues with the monetary policy framework (Plosser, 2015; Hilsenrath, 2016), I controlled for *dissent governor* and *dissent president* by counting the number of dissenting votes from governors and regional presidents at the FOMC meeting prior to each speech.

I also controlled for new information that was released into the market on the same day as the Fed speech. Based on prior work (Ederington and Lee, 1993; Nikkinen and Sahlström, 2004), I created dummy variables that were coded "1" if the *Consumer Price Index report*, the *Producer Price Index report*, and the *Unemployment report* were released on the same day as the speech. I also controlled for additional information the Fed introduced into the market. I created dummy variables that were coded "1" if *governor speeches*, *testimony*, and *press releases* were released on the same day as the speech.

Speech-related factors. I controlled for characteristics of the Fed speech itself that may correlate with my BR construct and influence market uncertainty. Consistent with research in financial economics (Spence, 1973; Van Buskirk, 2012), I controlled for the *speech word count*. To control for

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the level of *speech uncertainty*, I used the Financial Sentiments Dictionary created by Loughran and McDonaold (2011). Consistent with the idea that negative tone can offset positive tone (Pfarrer, Pollock, and Rindova, 2010), I controlled for *speech negative tone* using the LIWC software. Based on work that shows leaders exude a drive for power than influences markets (Winter, 1987; Emrich et al., 2001), I also used the LIWC software to control for *speech power*.

I controlled for "Fedspeak," which former Fed Vice Chair Alan Blinder (2001) refers to as complex, abstract, or vague language used by Fed Chairs to obfuscate sensitive subjects so as to avoid creating market uncertainty. I controlled for *speech complexity* by using the Flesch-Kincaid reading grade level (Kincaid et al., 1975); *speech abstractness* by using a word dictionary compiled and validated by Mergenthaler (1996); and *speech vagueness* by using a partial dictionary compiled and validated by Hiller and colleagues (1969).

Finally, I controlled for several general aspects of the Fed speech. I created a dummy variable for the *speech location* by assigning a "1" to speeches given in Washington, D.C., which is the Fed's headquarters and the meeting location of the FOMC. Consistent with work in finance (Chang, Pinegar, and Ravichandran, 1993; Dubois and Louvet, 1996; Choudhry, 2000; Connolly, Stivers, and Sun, 2005), I controlled for day on which the speech took place by creating dummy variables for each weekday. Based on interviews conducted with former Fed Chair Ben Bernanke and former Fed Governors Donald Kohn and Mark Olson, I also controlled for the topic of the speech and the local audience to whom the speech was delivered. Through an inductive analysis, nine *speech topics* were identified (i.e., state of the economy, financial crisis, financial literacy, central banking, banking system, globalization, economic history, commencement addresses, and remarks on special occasions) and five *local audiences* were identified (i.e., central bankers, banking industry, government, academics, and laypersons). A research assistant and I coded all speeches with these nine topics and five local audiences, with interrater agreement of 99 percent and 100 percent, respectively. I controlled for these two factors using dummy variables. To summarize, I employ the following regression model:

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$$\ln(VIX_t/VIX_{t-1}) = \alpha + \beta Backing\ Ratio_t + \eta Controls_t + \varepsilon_t$$

RESULTS

To conduct this event study analysis, I used OLS regression with year-fixed effects to estimate the effect of Fed speeches' BR on market uncertainty. Descriptive statistics and correlations for all major variables are shown in Table 1.

[Insert Table 1 about here]

Table 2 reports the results of the OLS regression models. Model 1 is the baseline model. Model 2 adds the BR as the primary independent variable of interest. Consistent with Hypothesis 1, I find that the more the Fed Chair explicitly reaffirms the backing underlying the monetary policy framework, the more this creates market uncertainty (see Figure 3). To interpret the practical significance of this effect, recall that the VIX is the expected range that the S&P 500 will move over the next month. A one standard deviation increase in the BR of a speech (i.e., 0.21)—which is equivalent to adding five backing-related paragraphs in a 25 paragraph speech—will increase this expected range by 1.0 percent. This effect is sizable in two respects. First, although the VIX is not directly tradeable, one can trade the VXX, which is an exchange-traded fund highly correlated with the VIX. A $10,000 investment would hypothetically gross a one-day return of $100 (1 percent). Second, portfolio managers and investors use the VIX to hedge their risk against market crashes (Rhoads, 2011). An increase in the VIX of even 1.0 percent makes this hedging strategy more expensive to execute, thereby influencing how market participants manage their long-term investment risk.

[Insert Table 2 and Figure 3 about here]

Model 3 adds the first interaction term between the BR and speech positive tone. Consistent with Hypothesis 2, I find that the positive tone of the speech suppresses the market uncertainty created from the Fed Chair explicitly reaffirming the backing (see Figure 4). This means that when a speech's positive tone is one standard deviation above its mean, the aforementioned 1.0 percent increase in the

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expected range of the S&P 500 over the next month goes away and, in fact, *reduces* the expected range by 0.5 percent. Interestingly, when a speech's positive tone is one standard deviation *below* its mean, this actually amplifies market uncertainty, increasing the expected range by 3.5 percent.

[Insert Figure 4 about here]

Model 4 adds the second interaction term between the BR and business media fear, and Model 5 reports the fully specified model. Consistent with Hypothesis 3, I find that business media fear present at the time of the speech amplifies the market uncertainty created from the Fed Chair explicitly reaffirming the backing (see Figure 5). This means that the aforementioned 1.0 increase in the expected range of the S&P 500 over the next month is amplified by 2.5 percent when the business media fear is one standard deviation above its mean. Moreover, similar to the complementary results of speech positive tone, when business media fear is one standard deviation *below* its mean, the aforementioned 1.0 percent increase market uncertainty goes away entirely.

[Insert Figure 5 about here]

Supplemental Analyses on the Financial Crisis Period

The theory developed in this paper seeks to explain the market consequences of a leader explicitly reaffirming the backing in a context where the underlying institutional assumptions are reasonably stable and taken-for-granted. This was generally true across the time period examined, since the monetary policy framework had not been fundamentally changed since 1977. However, it was at least slightly less true during the financial crisis. Indeed, this period created a substantial jolt that temporarily dislodged these assumptions from their taken-for-granted status. As a result, this supplemental analysis seeks to exploit this system jolt as a way to tentatively extend this story one step further by exploring what happens when the "high taken-for-grantedness" assumption of my theorizing is relaxed. In particular, when the taken-for-grantedness of these assumptions has already been weakened, which means that their objective and fact-like status has already been removed, explicitly

reaffirming these assumptions should no longer destabilize the institution to the same extent. By extension, this should not increase market uncertainty to the same extent as well.

To determine the appropriate financial crisis period, I leveraged the narrative timeline found in former Chair Ben Bernanke's memoir (Bernanke, 2015) that pointed to a period from 2008 to 2012. In January 2008, the Fed met for an unscheduled emergency FOMC meeting to reduce the federal funds rate by an enormous 75 basis points. This unprecedented action marked a key moment when the market recognized something was truly wrong. I concluded the period at the end of 2012 because the FOMC at this time uncharacteristically made clear how the Fed would be implementing their framework into the future. Specifically, they communicated that they would leave the federal funds rate exceptionally low at least through mid-2015, providing a bookend to much of the curiosity the market still had over how the Fed and their existing framework was handling the financial crisis.

I then ran Model 2 with an additional interaction term between the BR and a period dummy variable for the financial crisis period. The interaction term reaches marginal significance ($\beta = -0.051$, $\sigma^2 = 0.038$, $p = .088$, adjusted R-squared = 0.191). Note that this model includes both the financial crisis period dummy variable, along with year fixed-effects. Removing year fixed-effects produces a significant interaction ($\beta = -0.071$, $\sigma^2 = 0.036$, $p = .026$, adjusted R-squared = 0.122). If I split the sample and run Model 2 on speeches only given during the financial crisis period, the main effect results for H1 drop below significance ($\beta = 0.035$, $\sigma^2 = 0.036$, $p = .335$, adjusted R-squared = 0.244). This provides preliminary evidence of an important extension of my earlier theorizing. Specifically, to the extent the prevailing taken-for-grantedness of the institution has already been weakened, explicitly reaffirming these assumptions will not increase market uncertainty to the same extent.

To take this exploration one step further, I also consider how this weakened main effect might interact with the emotions present at the time of the speech. In particular, if this main effect is already diminished during the financial crisis period, then explicitly reaffirming the backing in the context of an especially optimistic configuration of emotions could potentially lead to an overall decrease (rather

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than increase) in market uncertainty. Specifically, this might occur for speeches given during the financial crisis period that also 1) contain high amounts of positive emotion and 2) occur when there is a low level of fear in the business media. To explore this, I further split the sample of speeches given during the financial crisis to consider only those that contained high positive tone (i.e., above the mean) and low business media fear (i.e., below the mean). I ran Model 2 again on this sample and, despite only containing 48 speeches, I found that the main effect entirely reversed (β = -0.155, σ^2 = 0.063, p = .022, adjusted R-squared = 0.363) (see Figure 6). What this suggests is that, when the taken-for-grantedness of the prevailing assumptions has been weakened and the emotional context is highly optimistic, explicitly reaffirming these assumptions actually *decreases* uncertainty.

[Insert Figure 6 about here]

Endogeneity Considerations

The primary endogeneity concern is that there is an omitted market variable that simultaneously explains why the Fed discussed the backing and why market uncertainty increased on the day of the speech. My research design attempted to address this concern in the most direct way possible by controlling for existing market uncertainty in the 30-days prior to the speech. I also controlled for other factors related to prevailing market conditions and included year fixed-effects. Most major market considerations should be reflected in at least one of these variables. Nevertheless, I seek to address endogeneity concerns in three additional ways to provide further comfort over the robustness of my results.

First, I conducted interviews with Fed officials to gain more insight into the factors influencing the timing of the speechwriting process and content of the speeches in my sample. Most notably, I interviewed Ben Bernanke (former Fed Chair from 2005 to 2014), Donald Kohn (Fed Governor from 2002 to 2010), and Mark Olson (Fed Governor from 2001 to 2006) (Kohn, 2017; Olson, 2017; Bernanke, 2017). The Fed schedules speeches roughly six months in advance, and selects speeches to fulfill commitments to local audiences and build relationships, not in response to market conditions.

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Speeches are written between one to six months in advance, depending on the type of speech. For speeches that discuss the current outlook of the economy (i.e., low BR speeches), writing starts a month before so the most current economic indicators are used. For speeches that reflect more upon the nature of monetary policy (i.e., high BR speeches), the writing process begins one to six months before. What this means is that the types of speeches that typically contain the most backing are written the earliest. Equally important is the fact that speeches rarely, if ever, are changed within one week of the speech date. When changes do occur, they are done primarily to update specific economic data (i.e., inflation rates, GDP, etc.). What this means is that the types of speeches most likely to be changed closer to the speech date are those least likely to contain high amounts of backing. This information is useful in that the more time that exists between when my independent variable changes (i.e., when the speech is written) and when my dependent variable changes (i.e., on the day the speech is delivered), the less likely it is that an omitted market variable can explain them both.

So what then makes the Fed Chairs talk about the backing? One key factor suggested to me by Bernanke during our interview is the local audience who invited the Chair to speak. While the Fed knows that the broader market pays attention to these speeches, the specific message is often written for the local audience, who differs substantially in the level of sophistication in regards to monetary policy (e.g., the Bank of England, an academic institution, an awards dinner, etc.). To examine this possibility, I correlated the BR with the five local audience dummy variables included in my analyses. From the most sophisticated local audience to the least, the results were central bankers ($r = 0.25$), banking industry ($r = 0.11$), government ($r = 0.00$), academics ($r = 0.01$), and laypersons ($r = -0.34$). A linear version of this variable correlates with the BR at 0.35, meaning that the more sophisticated the local audience, the more likely it is that the speech discusses the backing. This relationship is likely due to the fact that higher levels of sophistication are potentially required to understand the complexities and implications of the assumptions underlying the Fed's monetary policy framework. Note that this correlation is also higher than the correlation between the BR and any other market

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variable in Table 1. What this suggests is that the local audience does indeed appear to be a significant driver of whether or not the Fed discusses these assumptions. Equally important is that this also provides strong support for the construct and measurement validity of the BR, since it confirms the expectation of Bernanke, the author of many of the speeches in my sample. Nevertheless, note that controlling for the local audience does not change my results.

Second, I assess the likelihood that the two different types of market factors—acute and systemic—could operate as correlated omitted variables in my research design. Acute market factors are instantaneous injections of information or emotion into the market (e.g., press releases, earnings reports, etc.) and are the largest known drivers of my dependent variable. Indeed, daily movements in the VIX are extremely reactive to current events (Bennett, 2016), newly released information (Jamali, 2009), or emotional signals (Tetlock, 2007). A crucial observation is that since speeches are written between one to six months in advance, and high BR speeches are unlikely to be changed anywhere near the speech date, these acute market factors arising on or near the speech date are extremely unlikely to influence the speechwriting process months before. This suggests that while acute market factors are the most likely drivers of the change in my dependent variable, they are unlikely to affect my independent variable, thereby diminishing the likelihood that an acute market factor could be a correlated omitted variable.

Systemic market factors, in contrast, are prevailing market conditions that persist for some length of time and can influence daily market activities in a more indirect manner. Although I control for a number of systemic market variables, it is possible that certain factors are not reflected in these variables and may be driving my results. I conduct two tests to explore this possibility. The first test examines the expectation that if omitted systemic market factors were driving my results and, thus, driving the Fed to talk about the backing, then the BR of speeches given temporally near one another should be similar. Examining speeches given within three days of each another (N = 40 pairs) reveals an average difference in BR of 0.21 (or roughly one standard deviation), which is inconsistent with

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what one would expect if systemic market factors were driving the Fed to talk about the backing. The second test examines the possibility that a systemic market factor that ostensibly could be driving the BR of speeches is simultaneously affecting the underlying magnitude of the daily fluctuations in the VIX. Controlling for the previous average daily changes in the VIX (i.e., using 1, 10, or 30 days prior to the speech) did not change my results. Taken together, these observations diminish the likelihood that a systemic market factor could be a correlated omitted variable.

Third, I conducted an empirical test to assess how strong a correlated omitted variable would have to be to overturn my results. While this test is not new (Frank, 2000), it has only recently made its way into the management field (Hubbard, Christensen, and Graffin, 2017). I first calculated the Impact Threshold for a Confounding Variable (ITCV), which is the effect size the omitted variable would need to have in Model 2 to diminish my H1 results below significance. An omitted variable in my context would have to correlate *positively* with both the BR and market uncertainty at 0.12, which when multiplied together, makes my ITCV equal to 0.014. Figure 7 plots the ITCV along with the effect sizes of all other covariates in my model, calculated by multiplying each covariate's correlation with the BR with its correlation with market uncertainty. This figure shows that the size of the omitted variable needed to invalidate my results is substantially larger than every other control variable used in existing literature. Assuming that I have included a reasonable set of control variables, this suggests that my primary results are not likely driven by a correlated omitted variable.

[Insert Figure 7 about here]

Robustness Checks

I validated that my results were robust in a number of ways. First, I conducted all analyses again using generalized estimating equations (GEE) to control for the possibility of non-independence of these observations (Pfarrer et al., 2010; Wade, Porac, Pollock, & Graffin, 2006). Using the XTGEE routine in STATA 14.0, I chose the Gaussian (normal) distribution and an identity link function that corresponded to a linear model. For the correlation matrix, I assumed that there was a first-order

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autoregressive disturbance (AR1), in which disturbances from a Chair's prior speech are correlated with the disturbances resulting from the current speech. When doing so, I find strong support for all my hypotheses. Second, although the Chair generally talked about the backing in a descriptive and reaffirming manner, I coded each backing related paragraph for whether a potential change to the backing was discussed, the backing was described in future-oriented terms, and whether or not the backing appeared in the first five or ten paragraphs of a speech. Controlling for these factors again produced consistent results. Third, consistent with the expectation that the speech is not made public until the date and time denoted on the transcript, the BR did not predict market uncertainty for the two-day event window before the day of the speech ($\beta = .041$, $\sigma^2 = 0.017$, $p = .017$, adjusted R-squared = 0.083). Finally, I examined alternative specifications for a number of control variables to ensure that my results were robust to these changes. I used the actual federal funds rate instead of dummy variables, a two- and four-day window for the business media fear variable, and several different windows (e.g., 15-day and 150-day) for calculating the existing market uncertainty. My findings remained consistent across these specifications.

DISCUSSION

This study set out to investigate what happens when a prominent leader explicitly reaffirms the taken-for-granted assumptions underlying an institution. Using speeches made by the Chair of the United States Federal Reserve from 1998 to 2014, I proved that reaffirming the assumptions underlying the monetary policy framework causes uncertainty in financial markets around the world. Despite the robustness of this simple finding, it nevertheless feels at odds with the lingering intuition that reaffirming these assumptions should somehow reassert and strengthen their influence and, in turn, reinforce the existing social order. In fact, this is likely what these leaders were trying to accomplish. Why then do I find the exact opposite?

It is because the institutional assumptions being reaffirmed in my context had achieved a level of taken-for-grantedness that fundamentally altered their status and, in turn, the market's reaction to

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them being made explicit. When institutional assumptions are not highly taken-for-granted, they do not appear objective or fact-like and their contingent origins remain visible. Under such conditions, explicitly reaffirming these assumptions should reinforce and further institutionalize them. However, once these assumptions become taken-for-granted, and they take on an objective quality with their contingent origins obscured, explicitly reaffirming them will produce the opposite effect. Indeed, such efforts rip open the fact-like appearance of people's social world, exposing its underlying contingencies and creating uncertainty. In this sense, my findings are not in opposition to this original intuition, but rather, complementary to it in that together they portray a more holistic story about the relationship between language and institutionalization.

Consider the Fed over the last 40 years or so. In 1977, the Federal Reserve Reform Act created the new monetary policy framework that we all know today. During the years immediately following this institutional upheaval, this framework was not taken-for-granted by the market. While I cannot empirically validate this here, I would expect that explicit efforts to reaffirm these assumptions during this earlier period would have reinforced this framework further. By the 1990s, however, the assumptions that grounded this framework were largely taken-for-granted. It is under these conditions that I showed that the Fed Chair explicitly reaffirming these assumptions does not reinforce, but rather, disrupts the institution and creates uncertainty. Interestingly, as additional validation of this story, I also illustrated how, during the financial crisis, when the taken-for-grantedness of these assumptions had weakened, reaffirming them no longer creates uncertainty to the same extent. Taken together, my findings depict a consistent theory that demonstrates the close connection between reaffirming one's assumptions and the taken-for-grantedness underlying our social institutions.

This study, thus, demonstrates a simple yet potentially more general principle—*explicitly reaffirming assumptions that are true but taken-for-granted disrupts the natural course of things and creates uncertainty*. Indeed, we all hold assumptions that we take for granted and view as objectively true in a variety of everyday circumstances. For this reason, having a teacher, for instance, announce

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that "Today, I will not beat any of my students with a cane if they don't know an answer" does not reinforce what we believe to obviously be true, but instead, disrupts the natural course of things and suggests that things could be otherwise. Similarly, an airline pilot who reassures us when we board that "The safety checks that we just performed confirm that this plane is safe to fly" does not strengthen our confidence in the plane's safety, but rather, generates uncertainty by bringing to mind all the things that could go wrong but normally are not considered. The power of an institution's taken-for-grantedness, therefore, resides in the fact that these assumptions silently guide the way we think and act, enabling us to move on and consider other things. Explicitly reaffirming them at this point only disrupts the natural course of things, revealing the contingencies of our social world.

Beyond Market Signals

My primary finding also appears to be somewhat at odds with signaling theory (Akerlof, 1970; Spence, 1973), which is arguably the most dominant theory for understanding how communication influences financial markets (Tetlock, 2007; Graffin, Carpenter, and Boivie, 2011). According to this approach, information asymmetries exist between parties, which creates uncertainty. Prominent leaders, like the Fed Chair, can reduce this uncertainty either by sharing new information that market participants did not already know or by providing additional clarity to reduce noise so that market participants can interpret communications more easily (Connelly et al., 2011). In this context, reaffirming the assumptions that underlie the monetary policy framework likely does not provide new information, but rather, presumably seeks to clarify and increase the "signal to noise ratio" in the Fed's communications (Walsh, 2001; Blinder, 2009; Plosser, 2012), thereby intendedly reducing uncertainty in the market. This expectation is why, in fact, the Fed has taken such drastic steps over the last several decades to be more open and transparent in their communications (Yellen, 2013; Bernanke, 2015), and why my findings, which are inconsistent with this line of reasoning, are potentially counterintuitive to central bankers.

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Interestingly, I nevertheless demonstrate strong support for the signaling perspective in an entirely different way. Specifically, Table 2 indicates that longer speeches significantly reduce market uncertainty. Why do I find support for the signaling perspective in one sense but not another? The answer stems from how we theorize the market consequences of the different aspects of our communication. Longer speeches generally contain more information that market participants did not have beforehand. As a result, this new information should decrease information asymmetries between the Fed Chair and market participants, thus reducing uncertainty. However, not all aspects of these speeches contain new information. Although financial economists drawing on signaling theory tend to argue that communication that does not contain new information will produce no market effect at all (Tetlock, 2007), others suggest that non-informational reaffirmations of already existing knowledge can add clarity to people's interpretations and still reduce uncertainty (Blinder, 2009). However, the theory developed in this paper suggests that some efforts to add clarity in this manner can actually backfire. Indeed, when ideas have achieved a taken-for-granted status, reaffirming them only reveals the many contingencies of our social world, leading to the inadvertent creation of uncertainty. The theory developed here thus complements the signaling perspective by explaining the how more complex and socially embedded forms of communication operate in financial markets.

My findings also complement the signaling perspective in a second way, specifically related to the role of emotions in market contexts. Indeed, emotions are commonly viewed as signals (Tetlock, 2007) that contain a valence (e.g., positive or negative), conveying either more optimistic or pessimistic information to people and directly influencing the market's reaction. In contrast, this study portrays emotions not as a signal but as an interpretative context within which collective-level meaning making occurs. This portrayal is consistent with recent work by Pfarrer, Pollock, and Rindova (2010), who argue that celebrity firms have a positive emotional aura around them created by the media that influences how investors react to this firm when they announce earnings surprises. However, my findings extend their work in at least two important ways. First, I theorize and demonstrate that

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emotional contexts can be both positive and negative in valence, yet shape market participants' interpretation in similar ways. Indeed, a context created by high (low) levels of positivity shape people's interpretations in a similar way as a context created by low (high) levels of fear. Second, I also show that emotional contexts can be created not only from the media but also from the speaker. Indeed, speakers seem to be able to create an emotional context within which they can make potentially disruptive comments more safely. Thus, this study proposes a single theoretical mechanism to explain how two entirely different types and sources of emotion influence people's interpretations, providing some much needed insight into how collective-level emotions function in market contexts (Menges and Kilduff, 2015).

Transparency, Stability, and Hegemony

This study also raises important questions about the trade-offs between the growing trend towards public transparency and the stability of our markets, social institutions, and broader financial system. In particular, my findings demonstrate that one particularly powerful way leaders try to create transparency in their communication—by openly discussing the assumptions that ground their decisions and actions—can disrupt the very institution they intend to safeguard. Some have called this "the transparency paradox," where a well-intentioned push for greater transparency produces unintended consequences, prompting us to revisit the potential benefits of privacy (Bernstein, 2012). The Fed, of course, has struggled with this trade-off for decades. Historically, they preferred secrecy for fear that the markets would overreact to information that they do not understand. Yet over the last several decades, the Fed has become decidedly more transparent (Yellen, 2013; Bernanke, 2015). However, the challenge of balancing the ever increasing market demands for transparency against their goals for maintaining market stability is only growing. My findings highlight one specific way in which this trade-off manifests in everyday central banking communication.

This trade-off is not limited to central bankers either. In fact, these same issues confront most of our civic leaders, from Supreme Court Justices to Presidents, who regularly need to balance

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transparency with secrecy when communicating with their stakeholders. These considerations extend to corporate executives too, who regularly communicate with new investors (Lounsbury and Glynn, 2001; Martens, Jennings, and Jennings, 2007), securities analysts (Lee, 2015), other firms (Harmon, Kim, and Mayer, 2015), and employees (Rousseau and Tijoriwala, 1999), and presumably want to enhance clarity without inadvertently creating uncertainty. How and when should these leaders be transparent? My findings suggest at least a few preliminary guidelines. First, the specific way these leaders are transparent matters. For instance, executives might consider sharing more information or stories that stay "within the rules of the game," thereby increasing transparency in one sense while still silently reinforcing people's taken-for-granted assumptions. Second, leaders should also pay attention to the emotional context within which they communicate. Indeed, if they need to be more transparent about these taken-for-granted assumptions for whatever reason, they can ensure that discussing them does not create uncertainty by either encasing these statements within more positive language or postponing the discussion until a prevailing pessimistic climate passes.

These considerations, in turn, raise several questions about the relationship between public transparency and the stability of the broader financial system. In particular, since markets are culturally contingent (Fligstein and Calder, 2015), one can imagine that the optimal balance between transparency and secrecy might change if audiences take for granted different things (Lamin and Zaheer, 2012). More work needs to be done to better understand the effects of this audience heterogeneity embedded within increasingly complex financial systems (Greenwood et al., 2011). In addition, scholars have recently shown that firms can strategically use the taken-for-grantedness of categories as cover to take actions that avoid scrutiny (Hsu and Grodal, 2015). Do leaders use similar strategies when interacting with the market? This question raises important concerns about the subtle means by which powerful individuals exert control over our society (Gramsci, 1971). Specifically, if the existing institutional arrangement is not optimal but people take it for granted, are leaders

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deliberately not discussing these assumptions so as to reinforce the status quo? As Camaroff and Camaroff observe, "hegemony, at its most effective, is mute" (1991: 24).

However, research on the recent financial crisis (Lounsbury and Hirsch, 2010) suggests that such hegemonic concerns might actually be more complicated. This work proposes that the prominent leaders who oversee our financial system might themselves have become conditioned by their very own assumptions (Rubtsova et al., 2010; Marti and Gond, 2017). For instance, Fligstein, Brundage, and Schultz, after examining the Fed's FOMC transcripts, conclude that they "failed to anticipate the risks involved in the 2007 to 2008 financial crisis principally because of their overreliance on the frame provided by academic macroeconomics" (2017: 904). Abolafia echoes this same idea, suggesting that the Fed might get lost in the very game they themselves have created.

> The danger is that proficient masters of spin become so confident in their technical discourse that the restraints of uncertainty and legitimacy are no longer sufficient to encourage prudent questioning of the current operating models. The result is that discourse dominates the dialectic of technical rationality and the pragmatism of expert judgment is neglected…The threat is that the domination of any technical discourse inhibits the power of judgment to the point where the discourse is performing the policy rather the experts operating a discourse (2012: 112).

My findings, however, shed some doubt on the idea that the Fed was unaware of the framework within which they operated. Indeed, if they were blinded by their own assumptions, then they would not discuss the backing at all. The fact that the Fed seems to regularly discuss the backing, even during the financial crisis, may stem from the fact that I am one of the first to examine the Chair's personally developed speeches instead of the FOMC meeting transcripts. Indeed, FOMC meetings are held primarily to discuss the operations of monetary policy and, thus, unsurprisingly tend to contain mostly conversations that occur "within the rules of the game." In contrast, speeches are occasions where the Chair reflects more broadly about his or her beliefs regarding monetary policy operations as well as the framework itself. Importantly, even though the Fed was more aware of their own framework than prior

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work has assumed, central bankers have since acknowledged that they still do not know what they could have done to avoid a financial system collapse (PBS News Hour, 2008; Tarullo, 2017).

Taking these two observations together—that the Fed was indeed aware of their assumptions but that they did not know what could have been done to prevent the financial crisis—my findings point to an interesting suggestion. Perhaps our institutional leaders should discuss these taken-for-granted assumptions more regularly. Indeed, the sustained discussion of the underpinnings of our complex financial system would keep both leaders and market participants more aware of the assumptions we all are making. While doing so would create uncertainty, this is precisely the point. A healthy financial system over the long term requires a certain amount of uncertainty to deter people from taking extreme risks. This line of reasoning suggests that both researchers and practitioners might benefit from a better understanding of how we might reduce certainty (rather than uncertainty) in our markets and institutions (Zajac, 2013). Such efforts could provide some insight into how our awareness of these assumptions affects our decision-making (Beunza and Stark, 2012; Anthony, 2017), perhaps helping us develop alternative preventative measures to avoid future system-wide crises.

Micro-foundations of Institutions

This study also contributes to our understanding of the micro-foundations of institutions. Institutional theorists have conceptualized the micro-level plumbing of institutions in a variety of ways (Powell and Rerup, 2016), using frameworks based on social psychological evaluations or schemas (Bitektine, 2011; Tost, 2011; Glaser et al., 2016), emotions (Voronov and Vince, 2012; Creed et al., 2014), rituals (Dacin, Munir, and Tracey, 2010; Gray, Purdy, and Ansari, 2015), practices or performances (Zilber, 2002; Smets, Morris, and Greenwood, 2012; Glaser, 2017), and work (Lawrence and Suddaby, 2006; Lawrence, Suddaby, and Leca, 2009). Latent in most of these frameworks is the idea that language is somehow crucial to the way institutions are constructed, maintained, and changed over time, and a handful of scholars have prioritized studying how language itself influences these processes. For instance, Phillips, Lawrence, and Hardy (2004) have suggested that discourses are one

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way to understand how meaning coalesces in institutions, and Loewenstein, Ocasio, and Jones (2012) suggest vocabularies help us link micro-situational language usage with macro-collective meanings.

This study builds on these language-based perspectives to propose that *arguments* are a critical component of this micro-level linguistic plumbing that maintains and changes our institutions. While discourses and vocabularies reflect the system of words or statements actors typically use in different institutionalized contexts, arguments uniquely map onto the structural depth and taken-for-granted nature of our social institutions (Friedland and Alford, 1991; Jepperson, 1991; Thornton, Ocasio, and Lounsbury, 2012). This observation has enabled me to identify the *backing* as a conceptual and empirically observable aspect of our language that has a direct relationship with people's taken-for-granted assumptions and, by extension, the stability of our institutions. Indeed, leaving the backing implicit only strengthens people's taken-for-grantedness, thereby further stabilizing the foundations of the institution. In contrast, high levels of backing disrupts people's taken-for-grantedness, triggering mental alarms (Tost, 2011) or existential crises (Voronov and Vince, 2012), thereby jeopardizing the stability of the institution (Harmon, Green, and Goodnight, 2015). In this sense, the backing ratio represents a fulcrum between what we might think of as centripetal and centrifugal forms of public discourse that undergirds the stability and change of our institutions.

These observations also shed light on our understanding of cognitive legitimacy. While substantive forms of legitimacy (e.g., pragmatic and moral) are more easily observable because they reflect the presence of people's evaluation (Deephouse et al., 2017), cognitive legitimacy reflects how much actors take for granted certain ideas and, thus, actually represents the increasing absence of an evaluation itself (Suchman, 1995). Directly studying cognitive legitimacy and, more generally, the cognitive foundations of our institutions, has therefore been challenging when simply asking about such considerations draws unwanted attention to them. Green, Nohria, and Li (2009) have proposed that one option is to look for when justifications are no longer needed to convince others. Indeed, when we no longer need to justify a claim, this indicates that the justification has become implied in the

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overall line of reasoning, thereby representing a higher degree of cognitive legitimacy. The theory I have developed here complements this perspective by proposing the backing, or more specifically, the *backing ratio*, as a potential direct measure of cognitive legitimacy within a collective discourse. In particular, in the absence of overt pressures that silence people, a low backing ratio indicates that people do not feel the need to discuss, or call upon others to discuss, the assumptions that ground the institution, thus reflecting high levels of cognitive legitimacy. In contrast, a high backing ratio reflects low levels of cognitive legitimacy because it indicates that people feel the need to discuss, or request others to discuss, these assumptions directly. Arguments, therefore, uniquely reflect the underlying cognitive meaning structure of our institutions.

This perspective also provides a deeper understanding of some of the micro-level strategies actors engage in to maintain our institutions. In particular, my findings suggest that certain maintenance efforts, like rituals (Dacin, Munir, and Tracey, 2010) or storytelling (Zilber, 2009), successfully reproduce institutions, in part, because they do not make explicit these taken-for-granted assumptions. Indeed, such efforts powerfully reproduce the institution precisely because they indirectly reaffirm this taken-for-grantedness while still leaving its fact-like and objective status intact. In this sense, my findings highlight the potential unintentional consequences of when maintenance work is performed too explicitly. In this sense, my findings resonate with an enthnomethodological lens for exploring the micro-level interactions of institutions (Heaphy, 2013; Powell and Rerup, 2016). Indeed, a theory based on arguments, which simultaneously reflect people's "everyday reasoning" (Garfinkel, 1967) and the taken-for-granted structure of our institutions (Zucker, 1977), provides a strong conceptual basis for building a more robust micro-foundational theory of institutions.

Finally, while the primary thesis of this study concerns arguments and their influence on institutional processes, we should take care not to overlook the secondary thesis concerning the role of emotions. Indeed, a growing number of researchers are beginning to demonstrate the important micro-foundational role of emotions in shaping institutions (Voronov and Vince, 2012; Creed et al., 2014;

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Toubiana and Zietsma, 2017). In line with this work, I too have argued that emotions are not just an individual-level construct, but are also a collective-level force that can influence the institutional contexts in which people produce, maintain, and disrupt institutions. Yet how do these emotions change and from where do they originate? Emotions, of course, sometimes reside within institutions and people already. Yet this study suggests that language can also imbue a collectivity with emotion, keying them to new ways of thinking and interpreting (Goffman, 1967). Whether this keying originates from a prominent speaker or the media, language importantly can introduce a change in the emotional climate that can amplify and suppress the interpretation of potential institutional disruptions.

CONCLUSION

One of the classic sociological observations made by Berger and Luckmann (1967: 64) is that our "institutions are built upon language." Since that time, a number of theorists have referenced this theme (Friedland and Alford, 1991; Thornton, Ocasio, and Lounsbury, 2012) and have tried to explain what exactly they meant (Phillips, Lawrence, and Hardy, 2004; Suddaby and Greenwood, 2005; Green, Li, and Nohria, 2009; Loewenstein, Ocasio, and Jones, 2012). This study provides some clarity to Berger and Luckmann's abstract observation by demonstrating concretely that our linguistic assumptions—or what Toulmin calls the *backing*—is precisely what our institutions are built upon. By developing a perspective that connects arguments and emotions with the taken-for-grantedness of our collective understandings, this study generates new insights into the relationship between language and the micro-foundations of institutions.

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FIGURE 1 – The Toulmin Model

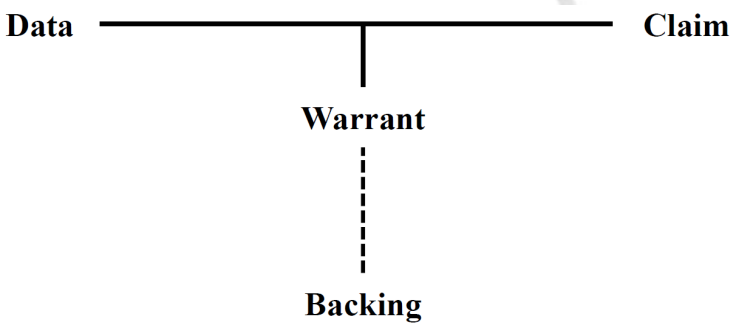


FIGURE 2 – Fed Chair Speeches from 1998 to 2014

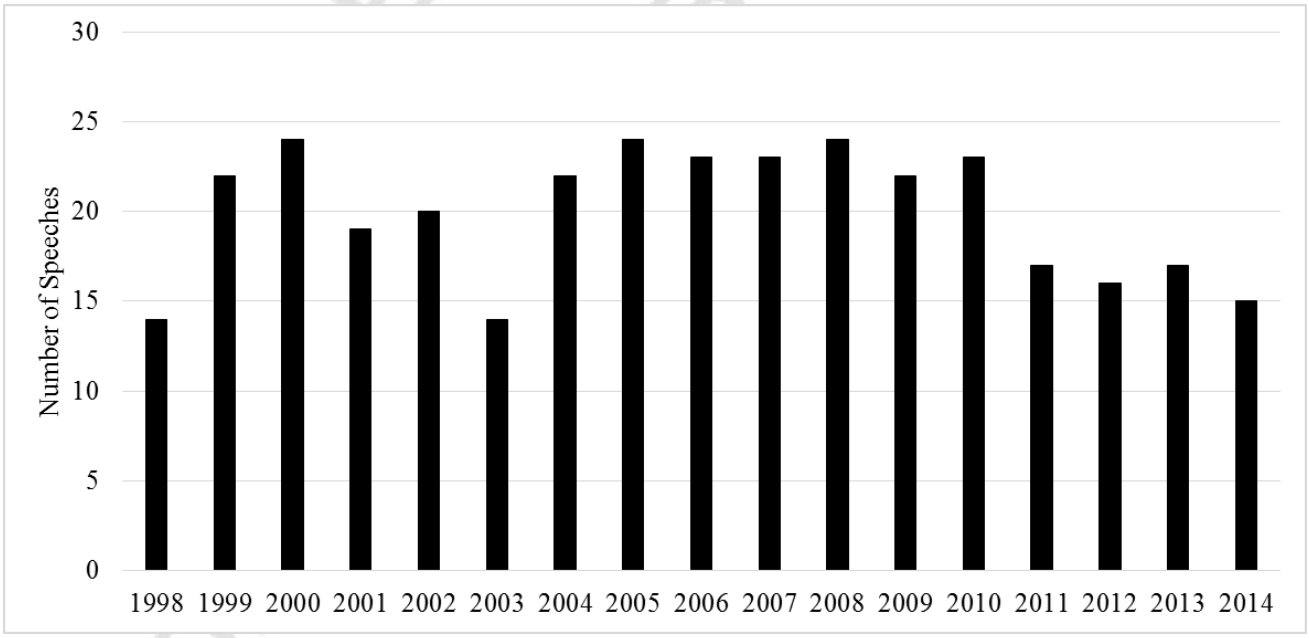


FIGURE 3 – Main Effect of Backing Ratio on Market Uncertainty

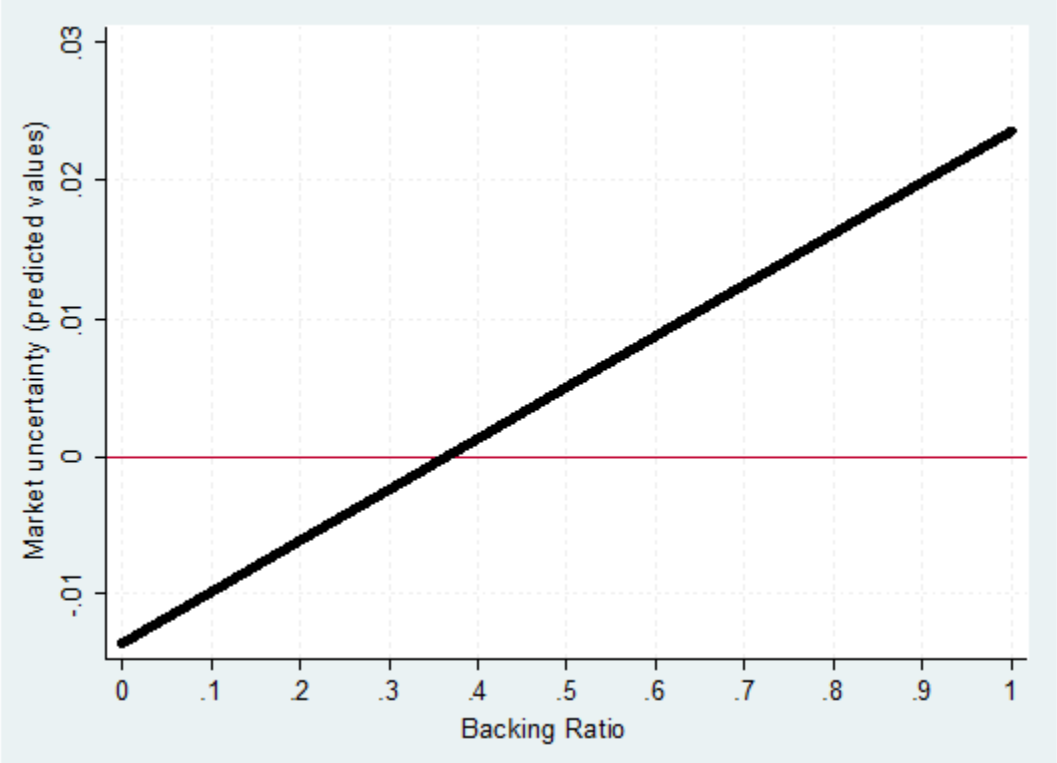


FIGURE 4 – Interaction between Backing Ratio and Speech Positive Tone

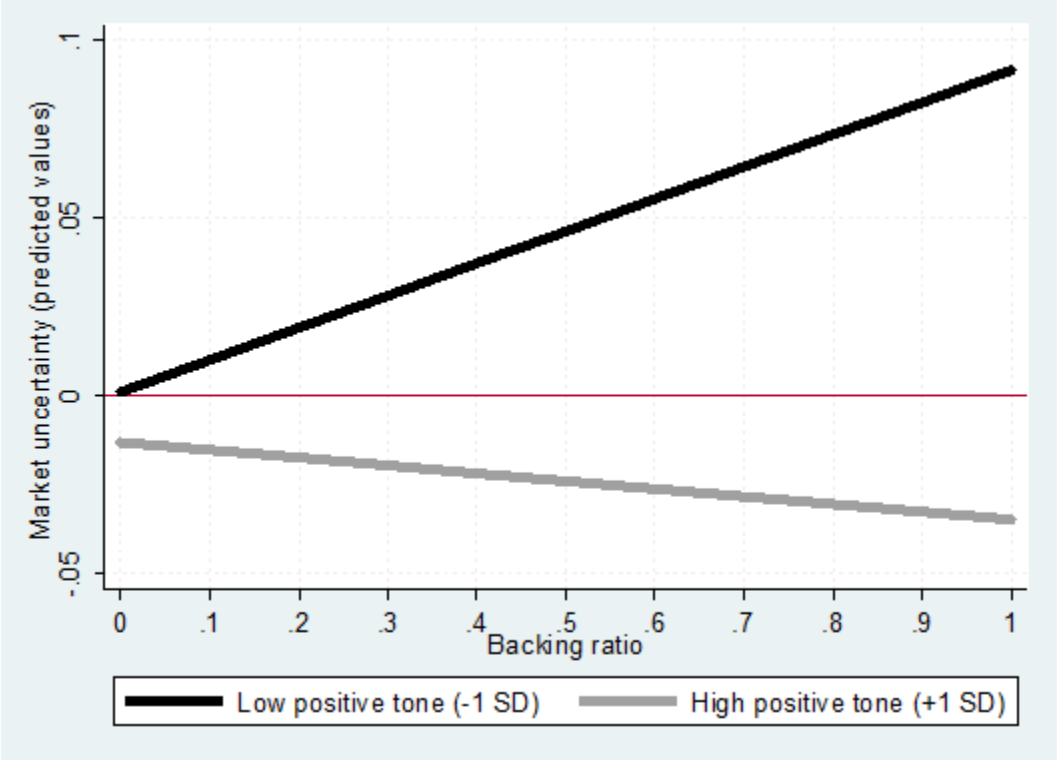


FIGURE 5 – Interaction between Backing Ratio and Business Media Fear

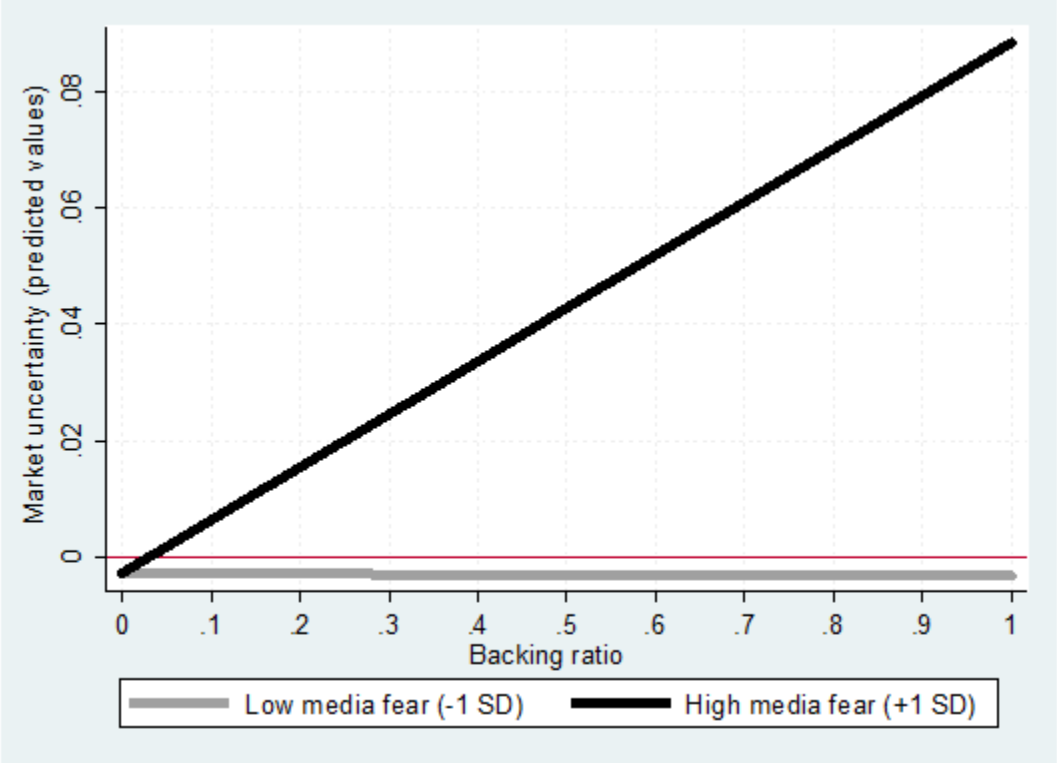


FIGURE 6 – Main Effect of Backing Ratio on Market Uncertainty during Financial Crisis
(Includes only speeches with *high positive tone*, given during times of *low business media fear*)

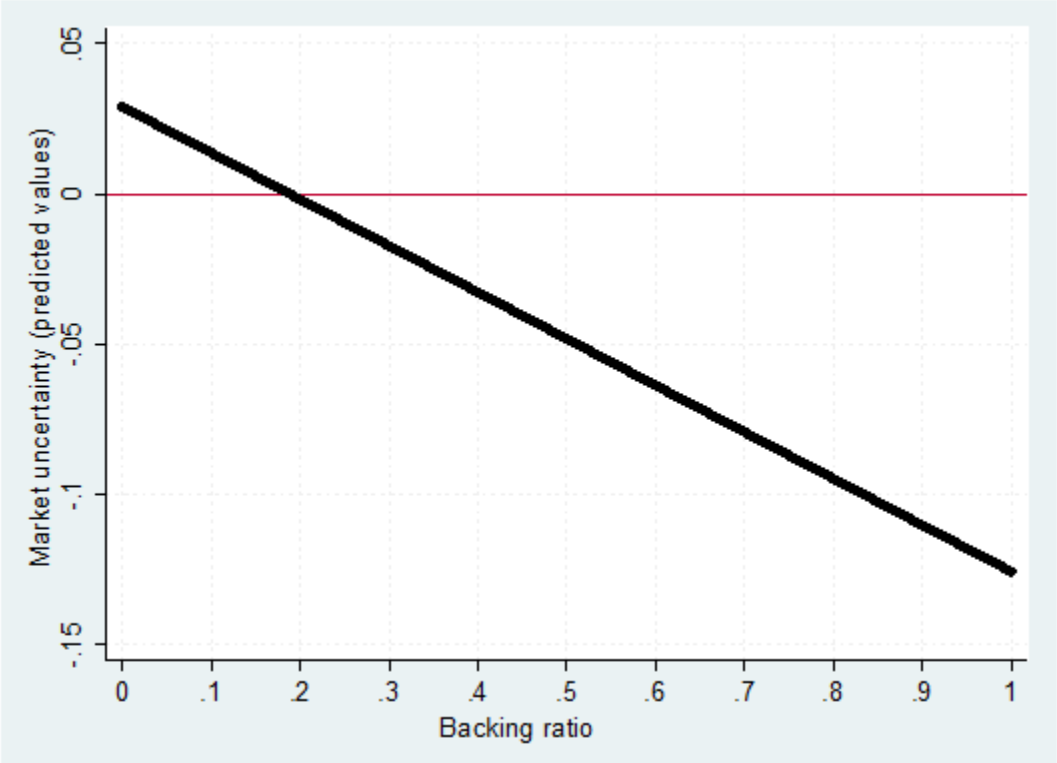
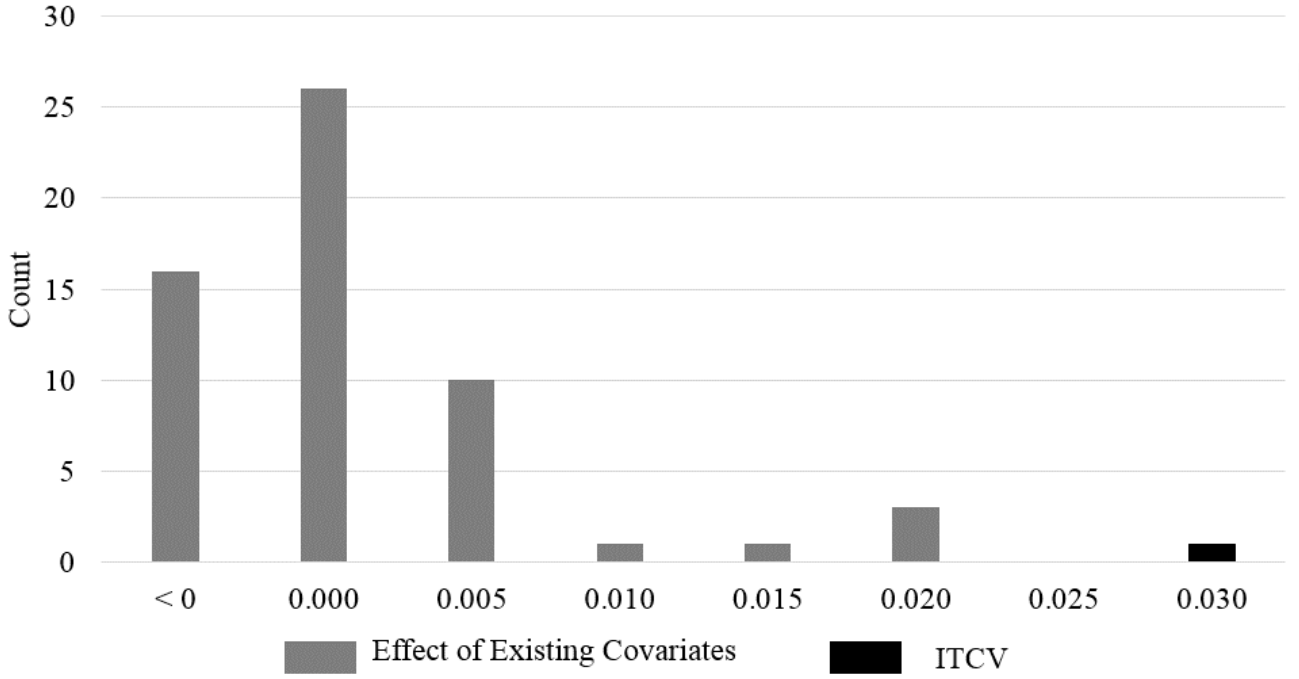


FIGURE 7 – Reference Distribution of the Effect Sizes for Covariates



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TABLE 1 – Descriptive Statistics and Pearson Correlation Statistics

		Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12
1.	Market uncertainty	-0.0051	0.06												
2.	BR	0.23	0.22	0.09											
3.	Speech positive tone	3.42	1.01	-0.05	-0.08										
4.	Business media fear	0.0084	0.00	0.10	-0.23	-0.17									
5.	Existing market uncertainty	21.84	8.83	-0.01	0.05	0.05	0.18								
6.	Unemployment rate	6.06	1.84	-0.08	0.26	0.22	-0.43	0.17							
7.	Inflation rate	2.36	1.27	0.10	-0.08	-0.03	0.30	-0.21	-0.48						
8.	Expansionary monetary policy	0.16	0.37	0.04	0.01	0.05	0.14	0.35	-0.21	0.24					
9.	Contractionary monetary policy	0.18	0.38	0.14	-0.17	-0.12	0.20	-0.28	-0.34	0.38	-0.20				
10.	Dissent governor	0.04	0.19	0.00	-0.10	0.00	0.13	0.06	-0.08	0.09	-0.08	0.16			
11.	Dissent president	0.47	0.64	-0.17	0.16	0.18	-0.23	-0.01	0.28	0.06	0.13	-0.28	0.01		
12.	Consumer Price Index report	0.07	0.25	0.06	0.08	0.03	0.03	-0.03	0.06	-0.02	-0.02	-0.03	-0.05	0.02	
13.	Producer Price Index report	0.06	0.24	0.14	0.05	-0.10	0.06	0.01	0.00	0.06	0.06	0.04	0.02	-0.04	-0.07
14.	Unemployment report	0.05	0.22	-0.09	0.05	-0.02	-0.06	0.04	-0.07	0.03	0.00	0.06	-0.05	-0.05	-0.06
15.	Governor speeches	0.23	0.42	-0.01	-0.03	-0.11	0.10	-0.01	-0.11	-0.03	0.05	0.08	0.01	-0.11	0.02
16.	Testimony	0.08	0.28	0.05	-0.02	0.12	0.05	0.06	0.02	-0.03	-0.01	0.00	0.06	0.01	-0.04
17.	Press releases	0.51	0.50	-0.06	0.06	0.07	-0.03	0.07	0.19	-0.03	0.04	-0.05	0.06	0.13	0.03
18.	Speech location	0.29	0.64	0.07	-0.10	0.18	-0.02	0.02	0.03	-0.06	0.02	-0.01	-0.05	-0.02	0.06
19.	Speech word count	2667.45	1210.28	-0.06	0.23	-0.41	-0.01	0.06	0.02	-0.07	0.03	-0.08	-0.09	-0.03	0.01
20.	Speech uncertainty	1.24	0.68	-0.01	0.08	-0.05	0.04	-0.08	0.04	0.01	-0.05	0.05	0.02	0.02	0.12
21.	Speech negative tone	1.74	0.86	0.08	0.27	-0.28	0.03	0.08	0.00	-0.04	0.04	0.01	0.01	-0.07	0.09
22.	Speech power	4.15	1.12	0.02	0.37	0.24	-0.20	0.02	0.23	-0.04	-0.02	-0.12	0.01	0.11	0.03
23.	Speech complexity	15.95	1.80	0.05	0.42	-0.27	-0.10	0.03	0.13	-0.03	0.00	-0.06	-0.05	0.09	0.04
24.	Speech abstractness	10.18	1.62	-0.16	0.03	0.04	0.01	-0.07	0.02	-0.07	-0.08	-0.04	-0.04	-0.02	-0.05
25.	Speech vagueness	1.12	0.37	-0.05	-0.08	-0.14	-0.08	-0.08	-0.01	0.03	0.05	-0.06	-0.05	0.04	0.03

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		Mean	SD	13	14	15	16	17	18	19	20	21	22	23	24
14.	Unemployment report	0.05	0.22	-0.06											
15.	Governor speeches	0.23	0.42	0.01	-0.07										
16.	Testimony	0.08	0.28	0.10	-0.02	-0.06									
17.	Press releases	0.51	0.50	0.03	-0.08	-0.03	0.06								
18.	Speech location	0.29	0.64	0.08	-0.09	-0.01	0.11	0.16							
19.	Speech word count	2667.45	1210.28	0.03	0.00	0.00	-0.11	-0.03	-0.30						
20.	Speech uncertainty	1.24	0.68	0.05	-0.07	0.07	0.05	-0.04	-0.03	0.11					
21.	Speech negative tone	1.74	0.86	0.11	-0.03	0.06	-0.12	-0.04	-0.19	0.42	0.07				
22.	Speech power	4.15	1.12	0.01	-0.05	-0.08	0.08	0.11	-0.01	0.03	0.04	0.11			
23.	Speech complexity	15.95	1.80	-0.02	0.00	-0.08	0.02	-0.05	-0.04	0.36	-0.01	0.16	0.21		
24.	Speech abstractness	10.18	1.62	0.07	-0.01	-0.01	-0.03	-0.03	-0.12	0.09	0.12	0.09	0.00	-0.06	
25.	Speech vagueness	1.12	0.37	-0.04	-0.06	-0.07	-0.06	0.01	-0.15	0.28	0.02	0.12	-0.06	-0.05	0.08

Note: 339 speeches from 1/1/1998 to 12/31/2014. Correlations above absolute value of 0.11 (0.14) are significant at 0.05 (0.01) for two-tailed test.

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TABLE 2 – Regression Models Predicting Market Uncertainty (t_{-1} to t_0)

	Model 1	Model 2	Model 3	Model 4	Model 5
BR		0.036***	0.034**	0.045***	0.041**
		(0.018)	(0.018)	(0.019)	(0.019)
Speech positive tone	-0.006*	-0.005	-0.007**	-0.005	-0.007**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Business media fear	-1.616	-1.176	-1.604	0.005**	-0.537
	(2.972)	(2.925)	(2.821)	(2.924)	(2.863)
BR * Speech positive tone			-0.056***		-0.049***
			(0.020)		(0.020)
BR * Business media fear				30.590***	26.293**
				(12.626)	(12.482)
Existing market uncertainty	0.000	0.000	0.000	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Unemployment rate	0.018	0.020*	0.021*	0.022**	0.022**
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Inflation rate	0.004	0.003	0.004	0.003	0.003
	(0.006)	(0.006)	(0.005)	(0.006)	(0.005)
Expansionary monetary policy	0.003	0.005	0.006	0.003	0.004
	(0.021)	(0.021)	(0.021)	(0.021)	(0.021)
Contractionary monetary policy	0.020	0.023*	0.025*	0.022*	0.025*
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
Dissent governor	0.009	0.011	0.007	0.017	0.013
	(0.019)	(0.019)	(0.018)	(0.019)	(0.018)
Dissent president	-0.020***	-0.020***	-0.018**	-0.017**	-0.015**
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Consumer Price Index report	0.019	0.017	0.020	0.019	0.021
	(0.015)	(0.015)	(0.014)	(0.014)	(0.013)
Produce Price Index report	0.028	0.026	0.028	0.026	0.028
	(0.018)	(0.018)	(0.018)	(0.017)	(0.017)
Unemployment report	-0.013	-0.015	-0.012	-0.012	-0.009
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
Governor speeches	-0.001	-0.002	-0.002	-0.001	-0.002
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Testimony	0.019	0.018	0.024**	0.022**	0.027**
	(0.011)	(0.012)	(0.011)	(0.011)	(0.011)
Press releases	-0.006	-0.006	-0.006	-0.005	-0.005
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Speech location	-0.002	-0.001	-0.001	-0.001	-0.001
	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
Speech word count	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Speech uncertainty	0.002	0.001	0.000	-0.000	-0.001
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Speech negative tone	0.009*	0.007	0.007	0.006	0.006

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	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Speech power	-0.002	-0.004	-0.005	-0.004	-0.005
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Speech complexity	0.001	0.000	0.000	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Speech abstraction	-0.006***	-0.006***	-0.007***	-0.006***	-0.007***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Speech vagueness	-0.005	-0.003	-0.004	-0.002	-0.003
	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)
Constant	-0.019	-0.023	-0.026	-0.044	-0.057
	(0.094)	(0.095)	(0.088)	(0.090)	(0.084)
Observations	339	339	339	339	339
R-squared	0.310	0.318	0.340	0.339	0.355
Adjusted R-squared	0.170	0.177	0.201	0.199	0.261
df	57	58	59	59	60
Average model VIF	4.10	4.11	4.09	4.09	4.07

Note: Results show robust regressions with robust standard errors in parentheses. Significance tests are one-tailed for directional hypotheses, two-tailed for control variables. All models include year, topic, and local audience fixed-effects.

* p<0.1, ** p<0.05, *** p<0.01

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